FED. SUP CLASS
5310

SELF-LOCKING NUTS SHALL BE SUBJECT TO THE FOLLOWING LIMITATIONS:

1. SELF-LOCKING NUTS SHALL NOT BE USED AS FOLLOWS:

   a. AT JOINTS IN CONTROL SYSTEMS AT SINGLE ATTACHMENTS OR WHERE LOSS OF THE BOLT WOULD AFFECT SAFETY OF FLIGHT UNLESS THE THREADED PARTS ARE HELD BY A POSITIVE LOCKING DEVICE THAT REQUIRES SHEARING OR RUPTURE OF MATERIALS BEFORE TORSIONAL LOADS WOULD RELIEVE THE INITIAL STRESSES OF THE ASSEMBLY.

   b. ON ANY EXTERNALLY THREADED PART THAT SERVES AS AN AXIS OF ROTATION FOR ANOTHER PART UNLESS THERE ARE NO POSSIBLE TORSIONAL LOADS WHICH CAN BE APPLIED TO EITHER THE EXTERNALLY OR INTERNALLY THREADED PART IN SUCH A MANNER AS TO RELIEVE THE INITIAL STRESSES OF THE ASSEMBLY, OR UNLESS THE THREADED PARTS ARE HELD BY A POSITIVE LOCKING DEVICE THAT REQUIRES SHEARING OR RUPTURE OF MATERIAL BEFORE TORSIONAL LOADS WOULD RELIEVE THE INITIAL STRESSES OF THE ASSEMBLY.

      EXAMPLE: PULLEYS, CRANKS, LEVERS, LINKAGES, HINGE PINS, AND CAM FOLLOWERS.

2. SELF-LOCKING NUTS MAY BE USED FOR THE MOUNTING OF ANTIFRICTION BEARINGS ON SHAFTS OR IN HOUSINGS WHERE THE NUT MEMBER SERVES TO CLAMP THE BEARING RING TIGHTLY AGAINST A SHOULDER OR TO ANOTHER MEMBER WHICH TRANSMITS THE CLAMPING LOAD TO A SHOULDER.

3. BOLTS, STUDS OR SCREWS MUST EXTEND THRU THE SELF-LOCKING NUT FOR A LENGTH EQUIVALENT OF TWO THREAD PITCHES. THIS LENGTH INCLUDES THE CHAMFER.

4. NUTS WHICH ARE ATTACHED TO THE STRUCTURE SHALL BE ATTACHED IN A POSITIVE MANNER TO ELIMINATE THE POSSIBLITY OF THEIR ROTATION OR MISALINEMENT WHEN TIGHTENING IS TO BE ACCOMPLISHED BY ROTATING THE BOLTS OR SCREWS. THE MANNER OF ATTACHMENT SHALL PERMIT REMOVAL WITHOUT INJURY TO THE STRUCTURE AND PERMIT REPLACEMENT OF THE NUTS. WHEN PROJECTION SPOT-WELDING IS USED FOR ATTACHING PLATE NUTS, CONTROL SHALL BE MAINTAINED IN ORDER THAT REMOVAL, BY DRILLING OUT THE WELDS PERMITS REPLACEMENT WITH DRILLED PLATE NUTS.

5. SELF-LOCKING NUTS THAT HAVE BEEN REWORKED OR REPROCESSED SHALL NOT BE USED BY CONTRACTORS OR FIELD MAINTENANCE PERSONNEL OF THE SERVICES.

6. SPECIAL NUTS, WHICH DEPEND ON FRICTION FOR THEIR ANCHORAGE AND TORSIONAL RIGIDITY, SUCH AS CLINCH NUTS, SINGLE-RIVET PLATE NUTS, AND SIMILAR DEVICES, ARE NOT ACCEPTABLE FOR USE IN AIRCRAFT STRUCTURAL APPLICATIONS. THEY MAY BE USED ON AIRCRAFT EQUIPMENT AND COMPONENT PARTS, SUCH AS INSTRUMENT MOUNTINGS AND ELECTRICAL EQUIPMENT.

7. SPLINE NUTS SHALL NOT BE USED IN AIRCRAFT STRUCTURAL APPLICATIONS BECAUSE OF LOW TORSIONAL STRENGTH AND MAINTENANCE DIFFICULTIES ENCOUNTERED IN REPAIRING OR REPLACING THEM.

8. SELF-LOCKING NUTS SHALL NOT BE USED IN CONJUNCTION WITH BOLTS OR SCREWS ON JET ENGINE AIRCRAFT IN LOCATIONS WHERE THE LOOSE NUT, BOLT, OR SCREW COULD FALL OR BE DRAWN INTO THE ENGINE AIR INTAKE SCOOP.

9. DO NOT USE SELF-LOCKING NUTS WITH BOLTS, SCREWS, OR STUDS TO ATTACH ACCESS PANELS, DOORS, OR TO ASSEMBLE ANY PARTS THAT ARE ROUTINELY DISASSEMBLED PRIOR TO OR AFTER FLIGHT.

10. DO NOT USE CADMIUM PLATED NUTS IN APPLICATIONS WHERE THE OPERATING TEMPERATURES EXCEED 450° F.

11. DO NOT USE SILVER PLATED NUTS IN CONTACT WITH TITANIUM AND TITANIUM ALLOY BOLTS, SCREWS OR STUDS IN APPLICATIONS WHERE THE OPERATING TEMPERATURES EXCEED 600° F.

12. SILVER PLATED NUTS SHALL NOT BE USED WITH SILVER PLATED BOLTS.

Ⓒ ENTIRE STANDARD REVISED

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

APPROVED 15 Mar 57 REVISED Ⓐ 23 Mar 60 Ⓑ 19 Apr 65 Ⓒ 27 Oct 67

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| P.A. Navy – AS<br>Other Cust<br>USAF – 11 | TITLE<br>NUTS, SELF-LOCKING, AIRCRAFT,<br>DESIGN AND USAGE LIMITATIONS OF | MILITARY STANDARD<br>MS33588(ASG) |
| PROCUREMENT SPECIFICATION<br>NONE | SUPERSEDES:<br>AND10068 | SHEET 1 OF 1 |

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